Modelling, Simulation and Digital Twin Exercises

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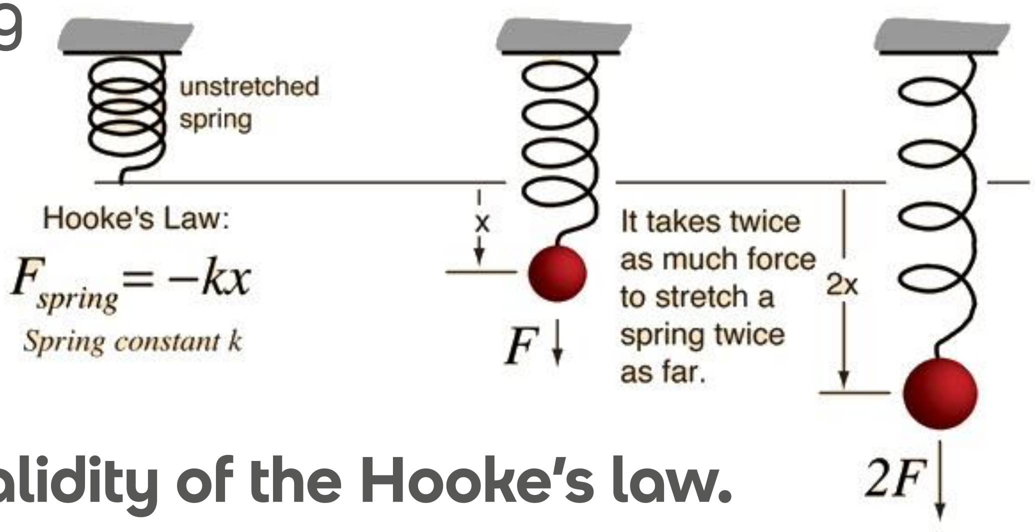
Exercise

01

Exercise 1

Physics-based modelling and empirical modelling

- Demonstrating physics via experiments
- Understanding trends within the data
- Fitting a curve to data



You design an experiment to demonstrate the validity of the Hooke's law.
The data collected are shown onto the next slide.

1. Plot the data. What relationship do you see between dependent and independent variables?
2. Fitting an appropriate curve to the data, can you make a prediction on the value of the spring constant in SI units?

[Hooke's law: https://phys.org/news/2015-02-law.html]

Force applied to the spring (N)	Displacement measured (mm)	Measurement uncertainty (mm)
0.1	7.3	0.5
0.2	11.1	0.5
0.3	22.4	0.5
0.4	25.5	0.5
0.5	32.0	0.5
0.6	46.0	0.5
0.7	46.2	0.5
0.8	52.9	0.5
0.9	61.0	0.5
1.0	60.2	0.5
1.1	72.9	0.5
1.2	84.3	0.5
1.3	87.7	0.5
1.4	96.6	0.5
1.5	99.8	0.5

[Further reading on data analysis using graphs: www.iit.edu/sites/default/files/2019-12/data_analysis.pdf]

Exercise

02

Exercise 2

The digital twin

In view of the learning from the first lecture, read the American Institute of Aeronautics and Astronautics white paper **Digital Twin: Definition and Value***

https://www.aiaa.org/docs/default-source/uploadedfiles/issues-and-advocacy/policy-papers/digital-twin-institute-position-paper-(december-2020).pdf

- What is in your view a digital twin?
- What is a use of the digital twin that you think could have value for a company?
- What are examples of products that could have a digital twin?

At the second lecture we will have an open discussion on these topics

*[*the language of the paper is quite technical. It is not important to understand everything, but just to have an idea]*

Exercise

03

Lecture 1 recap

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11. What is the least square method and how does it work?
12. What does extrapolation mean?

Exercise

04

P-diagram and Functional Flow Diagram

Create the P-diagram, the context diagram and the functional flow diagram for a product

It could be a complex engineering product or a simple product

This product could be used for the final assignment

Exercise

05

HEEDS Training

Install HEEDS (as per software guidelines on campusnet)

HEEDS documentation and Examples

https://www.egr.msu.edu/classes/me475/averillr/Lab1/HEEDS.pdf

Design of Experiments Example

Example 5: DoE Study of a Coil Spring

Optimisation Example

Example 10: Multi-objective optimisation

Exercise

06

Matlab Optimisation Toolbox

Introductory video:

https://de.mathworks.com/videos/optimization-toolbox-overview-70384.html

Key functions:

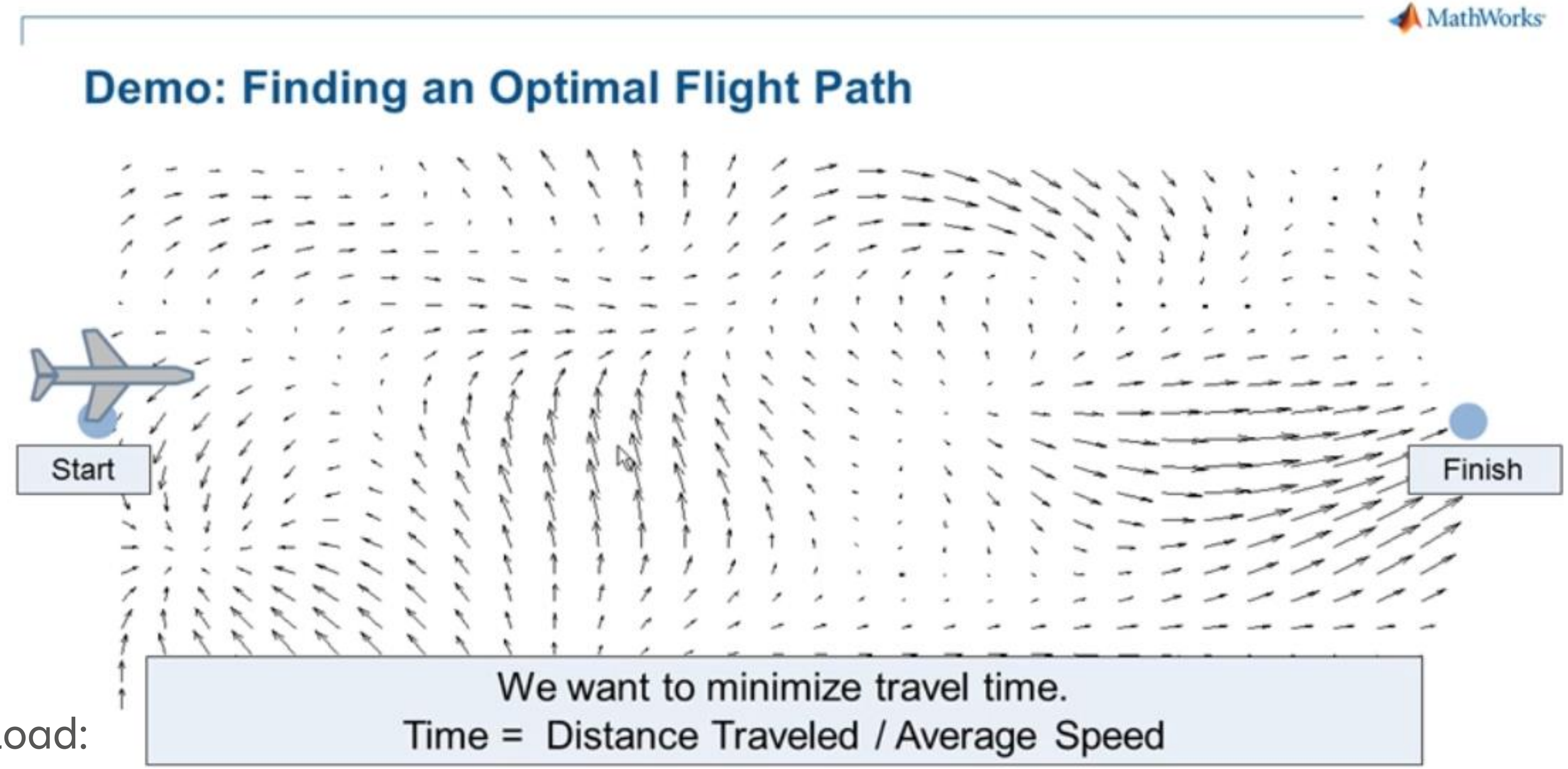
optimproblem, optimvar, optimconstr, optimexpr

Tutorial:

https://matlabacademy.mathworks.com/R2021a/portal.html?course=optim

Finding an optimal path

https://de.mathworks.com/videos/finding-optimal-path-using-optimization-toolbox-68958.html



Code download:

https://de.mathworks.com/matlabcentral/fileexchange/36321-finding-an-optimal-path

Exercise

07

Finite Element Analysis with Simcenter 3D

Simcenter 3D tutorial:

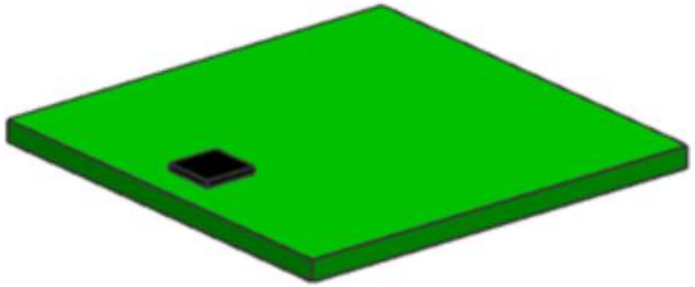
https://docs.plm.automation.siemens.com/tdoc/nx/1872/simcenter_3d_tutorials.html#uid:index_advanced_sim_tutorial

Exercise: **Thermal modelling - Analysing a Printed Circuit Board with a chip**

https://docs.plm.automation.siemens.com/tdoc/nx/1872/simcenter_3d_tutorials#uid:index_advanced_sim_tutorial:xid1688538

Additional Info:

https://docs.plm.automation.siemens.com/tdoc/nx/1899/nx_help/#uid:index



Simcenter Nastran Basic Tutorial (pdf):

https://docs.plm.automation.siemens.com/data_services/resources/scnastran/2020_1/help/tdoc/en_US/pdf/getting_started.pdf

Exercise

08

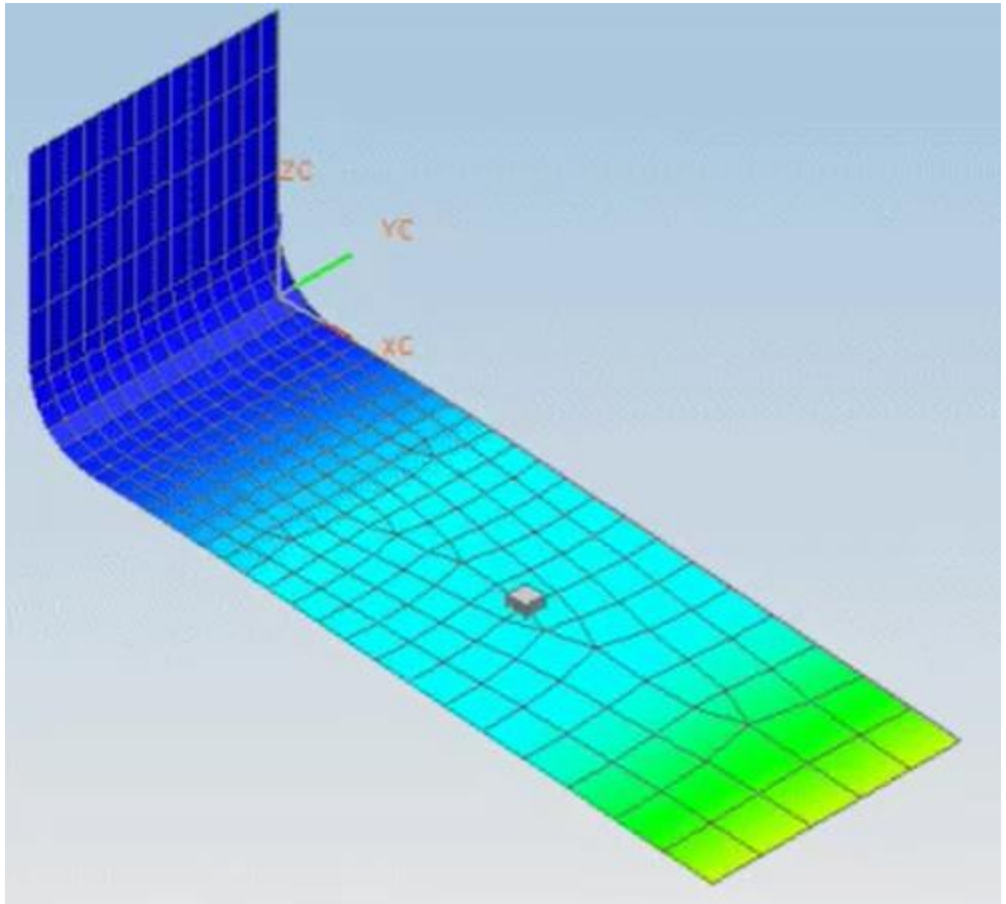
Finite Element Analysis with Simcenter 3D

Exercise: **Response Dynamics – Setting up the FEM**

https://docs.plm.automation.siemens.com/tdoc/nx/1872/simcenter_3d_tutorials#uid:index_advanced_sim_tutorial:xid1677563

Exercise: **Response Dynamics – Analyzing a transient event**

https://docs.plm.automation.siemens.com/tdoc/nx/1872/simcenter_3d_tutorials#uid:index_advanced_sim_tutorial:xid1677679



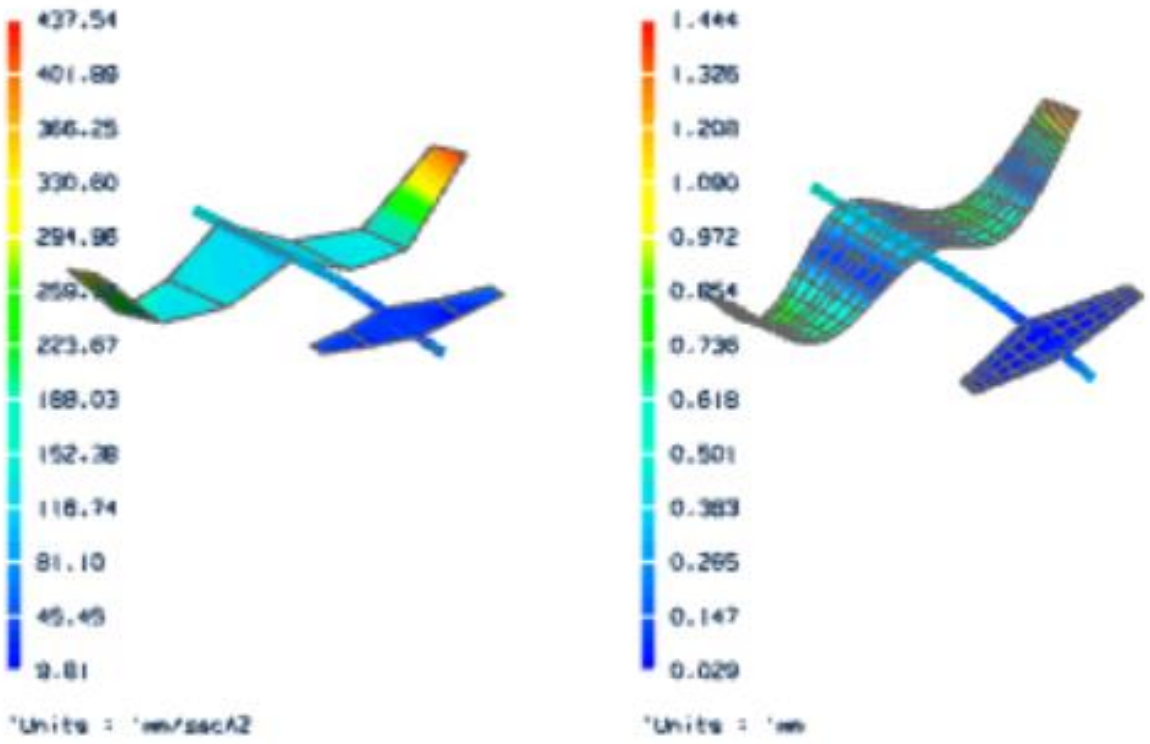
Exercise

09

Finite Element Analysis with Simcenter 3D

Exercise: correlate the results from a structural analysis solution with results from the test solution

https://docs.plm.automation.siemens.com/tdoc/nx/1872/simcenter_3d_tutorials#uid:index_advanced_sim_tutorial:xid1731225



Exercise

10

MATLAB Machine Learning Onramp

https://de.mathworks.com/learn/tutorials/machine-learning-onramp.html

Exercise

11

Diagnostics and Degradation Models with MATLAB

https://de.mathworks.com/help/predmaint/ug/wind-turbine-high-speed-bearing-prognosis.html#WindTurbineHighSpeedBearingPrognosisExample-1

https://de.mathworks.com/help/predmaint/ref/exponentialdegradationmodel.html

Exercise

12

Deep reinforcement learning and feature analysis

https://de.mathworks.com/matlabcentral/fileexchange/87939-playing-pong-with-deep-reinforcement-learning

https://de.mathworks.com/help/predmaint/ug/analyze-and-select-features-for-pump-diagnostics.html#AnalyzeAndSelectFeaturesForPumpExample-5

Vielen Dank für Ihre Aufmerksamkeit!

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